

Projective Dynamic Logo (PDL)

Global Mapping of Structures, Results, and Open Problems

Version 20 — A Guide for Continuation of the Programme

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Abstract

This document provides a systematic and self-contained guide to the Projective Dynamic Logo (PDL) programme, intended for a scientific colleague who wishes to understand, verify, or continue the research. PDL is a foundational physics programme that derives fundamental physical constants and dynamical laws from four axioms on finite signed graphs, without presupposing spacetime, particles, or fields. The minimal admissible closure under the axioms is the complete signed graph K_4 on four vertices and six edges, identified with the electron prototype. The proton is characterised by a unique integer quintuplet (24, 28, 930, 10087, 11017), from which all results flow.

The current corpus (D01–D53, plus auxiliary documents) has reached the following state:

- Three critical Gates have been resolved: the axiomatic derivation of the amplitude 155/11017 (Gate 1, D29), the structural proof that the QCD correction coefficient equals 2 (Gate 2, D30), and the theorem $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ (Gate 3, D31/D36).
- The quantum dynamics layer is **complete**: the Schrödinger equation (D32), Dirac equation (D33), Born’s rule Level 1 via Gleason uniqueness (D34), the Einstein coupling equation (D35), Born’s rule Level 2 including U(1) phase freedom (D46), and the London equation (D49) have all been derived from C1–C4 with no free parameter.
- Newton’s gravitational constant G is a theorem, modulo a single irreducible external parameter ($\Delta m_{\text{iso}} = m_d - m_u$).
- The Hubble tension is resolved at 0.006% with zero free parameters (D35/D27).
- Black hole thermodynamics is **complete**: the area law $S \propto R_{\text{surf}}$ (D37), the Bekenstein–Hawking formula $S_{\text{BH}}/k_B = 4\pi(M_{\text{eff}}/M_{\text{Pl}})^2$ (D38), the one-quarter coefficient $\frac{1}{4}$ in $S_{\text{BH}} = k_B A/(4\ell_P^2)$ (D50), and the London equation (D49) are all unconditional theorems.
- Open Problem OP1 has been fully resolved by D42: the Indifference Lemma H3 is now an unconditional theorem of axioms C1–C4.
- D46 resolves OP4: the U(1) phase freedom of quantum mechanics is derived from the K_4 pulsation structure, completing the quantum dynamics layer.
- D47 resolves OP13 and OP14: the spin–orbit splitting $\Delta n = 4$ is forced by a unique perfect-square discriminant (149^2), the sub-shell filling rates $r_{\text{exc}}(Z)$ are derived analytically for all $Z = 1\text{--}82$, the Mirror Lemma establishes proton–neutron HO-PDL level isomorphism, and the periodic table (valley of stability) is a corollary of C1–C4. The nuclear stability layer is **complete**.
- D48 partially resolves OP2-D35: the coherence stress-energy tensor C_{coh} is derived with three unconditional components ($C^{(\text{mass})}$, $C^{(\text{spin})}$, $C^{(\text{orb})}$) and one fully resolved component ($C^{(\text{leak})}$ via D51+D52).
- D49 resolves OP-London: the London equation $j_s^i = -(n_{\text{coh}}e^2/m_p c)A^i$ is derived from C1–C4 via the metric identity $\nu = \frac{1}{4}\|\psi_1 - T^k\psi_1\|^2$ (768/768 exhaustive verification) and axiom C4.

- D50 resolves OP12/BH-3: the coefficient $\frac{1}{4}$ in the Bekenstein–Hawking formula is derived from C1–C4 as the stable fraction $4/16$ per surface relation, under H3 (D42).
- D51 and D52 together resolve OP1-D35: the cosmological leakage constant is $C = (1 - \kappa)^{997} \times (930/11017)^{23} \times (1 - \eta_L)^{67}$, agreeing with the observational bound to 0.17 ppm. The causal chain $C1-C4 \rightarrow K_4 \rightarrow \beta_1 = 3 \rightarrow (p_{k_1}, p_{k_2}, p_{k_3}) \rightarrow C \rightarrow \Lambda_{\text{PDL}}$ is complete.
- D53 provides a self-contained synthesis of the complete causal chain $C1-C4 \rightarrow \Lambda_{\text{PDL}}$, with an independent numerical verification of C_{PDL} to 0.41 ppm relative to D51, and a systematic robustness analysis of the prime exponent structure. The chain spans fifty-seven orders of magnitude from combinatorial axioms to the cosmological constant.
- D40 derives the complete architecture of nuclear stability from the proton and neutron quintuplets alone: $N_{\min}(Z) = Z$ for $Z \leq 20$ as an exact theorem, the conflict saturation $C(Z > 20) = 190 T_{pp}$ exactly, and the valley of stability reproduced at 100% accuracy for all $Z = 1$ to 82.
- D41 confronts the PDL framework with the experimental results of Ha et al. (*Nature Communications* 16, 10631, 2025), confirming the structural identity $T \approx (\Delta n + 1)^2 = 25$ across three independent observables, with the wave-function component ratio agreeing at the 2% level.
- D43 establishes the computational infrastructure of the programme and proves the geometric leakage parameter $\varepsilon_{\text{geom}}(p) = 329/10087$ as an unconditional theorem (OP-A resolved).
- D44 derives the hierarchical filter factor $k = 0.921716$ from C1–C4 as an unconditional theorem (OP-B resolved). The complete causal chain $C1-C4 \rightarrow K_4 \rightarrow \text{quintuplet} \rightarrow \varepsilon_{\text{geom}} \rightarrow \varepsilon_G = \varepsilon_{\text{geom}} \times k^{18} \rightarrow G$ is now fully closed with no remaining open problem and no free parameter beyond Δm_{iso} .
- D45 derives the first falsifiable astrophysical confrontation: the primordial black hole survival threshold is shifted upward by +11.89% relative to the standard GR value, giving $M_{\text{PDL}}^* = \sigma(40)^{-2/3} M_{\text{GR}}^* \approx 5.706 \times 10^{14} \text{ g}$, testable with existing Fermi-LAT data via BLACKHAWK and ISATIS.

There is no longer any foundational open problem at the axiomatic level. The causal chain from C1–C4 to the cosmological constant Λ_{PDL} is complete. The remaining open problems (OP2–OP17 and those inherited from D48) concern higher-level derivations, extensions, and experimental confrontations.

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1 Purpose and Scope

1.1 Aim of this document

This mapping document pursues four objectives. First, it provides a complete and current inventory of the PDL corpus (D01–D48, plus auxiliary documents), with concise descriptions of each document’s role and main result. Second, it makes explicit the logical dependency structure of the programme: which results follow from which, what has been proved, and what remains conjectural or open. Third, it classifies all results according to a rigorous epistemic taxonomy, distinguishing proved theorems, conditional theorems whose hypotheses are explicitly named, structurally motivated conjectures, and open problems. Fourth, it identifies the most productive entry points and continuation paths for a colleague wishing to extend the programme. This document is not a replacement for the primary technical documents; it is a navigational and epistemic guide that complements them.

1.2 What makes this version different from v19

Version 19 of the mapping document covered the corpus up to D52 and marked the complete closure of the causal chain $C1-C4 \rightarrow \Lambda_{\text{PDL}}$. The present version v20 incorporates one additional document (D53) and completes the dissemination layer of the cosmological leakage result.

- D53 provides a self-contained synthesis of the causal chain $C1-C4 \rightarrow \Lambda_{\text{PDL}}$, presenting all steps from D51 and D52 in a single linear text. Its primary contribution is an independent numerical verification at 50-decimal precision (three fully self-contained Python cells) confirming C_{PDL} to 0.41 ppm relative to the value published in D51. The robustness analysis confirms that replacing any prime exponent by its nearest composite neighbour degrades the agreement by at least four orders of magnitude. D53 also provides the diagnostic that the apparent 8.17 ppm deviation relative to Planck 2020 is entirely metrological in origin (8.6 ppm offset in $(m_p c/\hbar)^2$ between the CODATA versions used) and carries no structural significance.

The reader familiar with v19 should note the following changes: D53 is added to the corpus table; the reading guide gains an entry for D53; the continuation guide references D53 as the recommended entry point for the cosmological constant derivation; and the bibliography gains one entry.

1.3 Epistemic conventions

Throughout this document, the following colour-coded epistemic labels are used in tables and text:

Theorem A result established by a complete proof from the PDL axioms C1–C4 and previously proved theorems, with no free parameters and no unproved auxiliary hypotheses.

Theorem (Hx) A result proved under one or more explicitly named hypotheses H1, H2, H3. . . ; the proof itself is complete conditional on those hypotheses.

Conjecture A structurally motivated claim supported by numerical evidence, qualitative arguments, or partial proofs, but not yet established rigorously.

Named Hypothesis An auxiliary assumption that is explicitly labelled and isolated, so that results depending on it are clearly traceable.

Open Problem

A well-posed question whose resolution would advance the programme significantly.

2 Document Corpus

2.1 Complete document table

Table 1 lists the complete PDL corpus in Zenodo canonical order.

Table 1 PDL corpus in Zenodo canonical order. *Main result* denotes the single most consequential contribution of each document to the programme chain.

Label	Title (abbreviated)	Main result / role
D01	The Emergence of Physical Reality within the PDL Framework [1]	Founding article: four relational axioms, structure, proton architecture, first treatment of α , μ , G , and cosmology.
D02	Introduction to the PDL [2]	High-level conceptual overview; standard first reading for new collaborators.
D01F	L'émergence de la réalité physique (LDP FR) [3]	French translation of D01.
D03	The Emergence of Physical Reality (IM-RaD format) [4]	Journal-formatted version of D01.
D04	A Modal Dialogue between PDL and the Theory of Objectivity [5]	Comparative study of PDL and TO; ontological commitments of PDL.
D05	A Minimal Relational Sketch for the Golden Ratio in PDL [6]	Self-similar core-surface partition; golden ratio and active surface $(\sigma, 3)$.
D06	Hierarchical Filtering of Coherence Leakage and the Exponent 18 [7]	First formulation of the 18-filter leakage hierarchy.
D07	Gleason-Type Uniqueness of Born's Rule for Spin- $\frac{1}{2}$ [8]	Early sketch of the Gleason approach; superseded by D34 at Level 1.
D08	Logical Leakage as Self-Maintained Probability: Topological Reformulation [9]	Coexistence topology; coherence-cost pseudometric; leakage as probability.
D09	PDL as a Foundational Research Programme [10]	Programme position paper; first systematic open-problem list.
D10	From Discrete Coherence Flux to Effective Fields [11]	Gauss- and Faraday-like relations from coherence fluxes; path towards Schrödinger.

Label	Title (abbreviated)	Main result / role
D10a	Proper Time as Coherence-Cycle Counting [12]	Proper time as a count of coherence cycles; emergent Minkowski structure; $\omega_p = m_p c^2 / \hbar$ forces $\lambda_C = \hbar / (m_p c)$ and $V_C = (4/3)\pi\lambda_C^3$ (chain confirmed in D48).
D11	Towards an Einstein–Dirac Unification from PDL [13]	Preliminary sketch coupling Dirac spinors to the emergent coherence-cost metric.
D12	A Structural Derivation of the Fine-Structure Constant from PDL [14]	Closed-form expression for α from $(930, \sigma, \varphi)$; accuracy $\mathcal{O}(10^{-4})$.
D13	Compatibility of the Schrödinger Framework with PDL [15]	Benchmark: PDL-derived α and μ in hydrogenic Schrödinger; spectral concordance.
D14	Born’s Rule and the Golden-Ratio Active Surface in PDL [16]	Relational derivation of Born’s rule for spin- $\frac{1}{2}$; role of φ in σ .
D15	Towards a Schrödinger-Type Dynamics from PDL: A Relational Sketch [17]	Early sketch of Schrödinger emergence; superseded by D32.
D16	Combinatorial Selection of the Proton Architecture: Uniqueness Conjecture [18]	Optimisation functional for the quintuplet; local uniqueness conjecture.
D16a	Minimal Stationary Closures in PDL: Necessity of the $(4, 6)$ Block [19]	Theorem: no admissible closure exists on $ V < 3$; K_4 is first admissible.
D16b	Combinatorial Selection and Local Uniqueness of the Proton Architecture [20]	Refined combinatorial treatment; local uniqueness conjecture for $(24, 28, 930, 10087, 11017)$.
D17	Coherence Leakage, Hierarchical Filtering, and the Exponent 18 [21]	Organisation of the 18 filters into three blocks; effective coupling ansatz ε^{18} .
D18	Discrete Cavity Modes and Logical Density of States [22]	PDL cavity modes; $2n^2$ density-of-states in three dimensions.
D19	Existence as Pulsating Closure: An Ontological Meditation [23]	Philosophical analysis of PDL boundaries, leakage, and transcendence.
D20F	Qui que nous puissions être (FR) [24]	French philosophical synthesis; companion to D20.
D20	Whoever We May Be: A Philosophical Synthesis of PDL [25]	Ontological and existential synthesis; accessible entry point for non-specialists.

Label	Title (abbreviated)	Main result / role
D21	Universal Coherence Leakage: A Structural Bridge between G and α [26]	Primary result: $R_{\text{tot}}/R_{\text{surf}} = 2\nu^2/(2\nu^2 - 1)$; $G \approx 6.67448 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$ (27 ppm vs. CODATA 2022).
DN	<i>Whatever We May Be</i> : The PDL popular book [27]	Popular science monograph; non-technical introduction to PDL.
D22	Nuclear Stability and the Periodic Table as Combinatorial Closure Hierarchies [28]	Neutron quintuplet (24, 28, 1032, 9960, 10992); $N_{\text{crit,max}} = 126.1$; $Z_{\text{sat}} = 20$; Mirror Lemma (confirmed analytically in D47).
DM	PDL Global Mapping v20 [29]	Navigation document for the complete corpus D01–D53; the present document.
D23	Topological Origin of the Exponent 18, S^2 , and the Periodic Table [30]	Theorem: $18 = 6 + 5 + 4 + 3$ is topologically necessary via S^2 ; $n = 3$ forced by S^2 topology (confirmed in D48 as second independent chain for V_C).
D24	Closure-Density Dependence of κ and the Hubble Tension [31]	Introduction of $\sigma(N) = 1 - (1 - \kappa)^N$; structural origin of H_0 discrepancy.
D25	A Parameter-Free Structural Bridge between α and G [32]	Primary result: full bridge formula; G at 27 ppm; $\kappa \approx 0.0075194$.
D26	Cosmological Resolution via PDL: G as Topology-Dependent [33]	G as topology-dependent parameter; Hubble tension as scale-transition signature.
D27	Structural Derivation of N_{CMB} from the PDL Neutron Architecture [34]	$N_{\text{CMB}} = 40$ from n_6 ; $H_{0,\text{CMB}} = 67.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$; 0.27σ Planck.
D28	The PDL–QCD Boundary: Mass Ratio, Correction, and the ε_G^B Conjecture [35]	$\mu = \frac{155}{11017}(1 - 2\kappa^2)\frac{m_{\text{iso}}}{m_p}$; ε_G^B conjecture formulated.
D29	Axiomatic Derivation of 155/11017: Gate 1 Resolved [36]	Theorem: 155/11017 proved from C1–C4 via AB; 768/768 exhaustive verification.
D30	QCD Correction Coefficient $a = 2$: Gate 2 Resolved [37]	Theorem: $a = 2$ structurally forced; $m_{\text{iso}} = m_d - m_u$ is the unique irreducible external parameter.
D31	Proof of $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$: Gate 3 (preliminary) [38]	First proof of Gate 3 under linear bridge inversion hypothesis; superseded in rigour by D36.
D32	Schrödinger Dynamics from the AB Coupling Criterion [39]	Theorem: Schrödinger equation derived; $b = -i\hbar/(2m_e)$; ∂_x^2 forced by AB.
D33	Dirac Equation from the $\text{SL}(2, \mathbb{C})$ Pulsation of K_4 [40]	Theorem: $T = -i\sigma_2$ unique among 8 candidates; $T^2 = -I_2$; Clifford algebra; NR limit = D32 exactly.

Label	Title (abbreviated)	Main result / role
D34	Born Rule from AB-Admissible Amplitudes [41]	Theorem Level 1: five Gleason axioms verified; Born rule unique from Gleason uniqueness.
D35	Quantitative Einstein Equation from $\sigma(N)$ [42]	Theorem (now unconditional via D42): $G_{\text{eff}}(N) = G_{\text{Newton}}$; Hubble ratio at 0.006%; synthesis document.
D36	Proof of $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ from the Trace Structure of K_4 [43]	Theorem (now unconditional via D42): $\sigma_0 = 0$ exact from AB; $\text{Tr}(A) = 0$ proved; Gate 3 strengthened.
D37	Surface Locality of Relational Entropy and the PDL Area Law [44]	Theorem BH-1 : entropy of a PDL closure is entirely carried by R_{surf} ; area law $S \propto R_{\text{surf}}$ unconditional.
D38	Bekenstein–Hawking Entropy from the PDL Effective Gravitational Coupling [45]	Theorem BH-2 (now unconditional via D42): $S_{\text{BH}}/k_B = 4\pi(M_{\text{eff}}/M_{\text{Pl}})^2$; verified to 10^{-15} ; three falsifiable PBH predictions.
D39	Derivation of κ from the PDL Axioms: the Indifference Lemma [46]	Theorem (H3) : $\kappa = 3/\sigma$ under H3; partial resolution of OP1; $D36 \Rightarrow H3$ is the unique residual; H3 now a theorem (D42).
D40	Nuclear Stability and the Periodic Table from PDL Combinatorial Axioms [47]	Theorems : $N_{\min}(Z \leq 20) = Z$ exact; $C(Z > 20) = 190 T_{pp}$ exact; valley of stability at 100% for $Z = 1\text{--}82$; eight assembly rules; connection identified between OP14 and OP12.
D41	Structural Closure Multiplicity and the Isospin-Symmetric Island of Inversion [48]	First PDL confrontation with nuclear spectroscopy data (Ha et al. 2025). Three independent observables consistent with $T = (\Delta n + 1)^2 = 25$: BE2 ratio (22%); N_{comp} ratio (2%); ΔS_{WF} (25%). Conjecture H_B introduced; falsifiable predictions P7 and P8.
D42	Derivation of H3 from Axioms C1–C4: The Equiparticipation Lemma [49]	Theorem : H3 derived unconditionally via D42-L1 (Equiparticipation), D42-L2 (cross-triangle blindness), D42-L3 (S_4 -equivariance; 24,576 cases). Corollary: $\kappa = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$ unconditional. OP1 fully resolved.
D43	Numerical Verification of the PDL Causal Chain: 18-Filter Architecture and Colab Infrastructure [50]	Systematic numerical verification at 50-decimal precision. Proves $\varepsilon_{\text{geom}}(p) = 329/10087$ and $\varepsilon_{\text{geom}}(n) = 468/9960$ as unconditional theorems (OP-A resolved). Reusable Colab infrastructure.
D44	Closure of OP-B: Derivation of the Hierarchical Filter Factor k from the PDL Axioms [51]	Theorem : $k = 0.921716$ derived from C1–C4 unconditionally. OP-B resolved; OP8 resolved; complete causal chain $C1\text{--}C4 \rightarrow G$ closed.

Label	Title (abbreviated)	Main result / role
D45	Primordial Black Hole Evaporation Threshold from PDL: A Falsifiable Prediction for Fermi-LAT [52]	First falsifiable astrophysical confrontation. $M_{\text{PDL}}^* = \sigma(40)^{-2/3} M_{\text{GR}}^* \approx 5.706 \times 10^{14} \text{ g}$ (+11.89%). Excess Hawking flux at $E_{\text{peak}} \approx 90\text{--}104 \text{ MeV}$ in IGRB; test protocol via BLACKHAWK + ISATIS + Fermi-LAT.
D46	Born's Rule Level 2: Derivation of U(1) Phase Freedom from the K_4 Pulsation [53]	Theorem OP4 resolved: Proposition 1 — K_4 sign equivalence $s \sim -s$ and $T^2 = -I_2$ generate a canonical U(1) action with Hopf fibration $S^1 \hookrightarrow S^3 \rightarrow S^2$. Proposition 2 — $(A) \wedge (B)$ criterion is U(1)-invariant (verified over all 8 configurations). Proposition 3 — unique U(1)-equivariant measure on S^3 satisfying Level 1 Gleason axioms is Born's rule. Quantum dynamics layer complete. Connection to Berry phase and London/Meissner analogy via $C^{(\text{orb})}$ (D48) documented.
D47	Derivation of Sub-Shell Filling Rates and the Periodic Table from the PDL Axioms: Resolution of OP13 and OP14 [54]	Theorem OP13 resolved: $\Delta n = 4$ is the unique value in $\{0, 4, 8, \dots\}$ such that the discriminant of the quasi-completeness equation is a perfect square (149^2), forcing $n_u = 24$, $n_d = 28$, $s = 1/12$ unconditionally. Mirror Lemma (Lemma 5.2): proton and neutron HO-PDL levels are isomorphic, 20 levels verified. Theorem OP14 resolved: $r_{\text{exc}}(Z) = 0$ for $Z \in \{28, 50, 82\}$ and $r_{\text{exc}}(Z) = 1$ otherwise, 31/31 sub-shell boundaries, zero free parameter. Corollary: $N_{\text{min}}(Z)$ for $Z = 1\text{--}82$ is an unconditional theorem of C1–C4. Nuclear stability layer complete.
D48	Derivation of the Coherence Stress-Energy Tensor from the PDL Axioms: Explicit Form of C_{coh} and Partial Resolution of OP2-D35 [55]	Theorem OP2-D35 partially resolved. $V_C = (4/3)\pi[\hbar/(m_p c)]^3$ doubly forced (D10a temporal chain + D23 geometric chain), $V_{\text{geom}}/V_C = 1.00000000$ at machine precision. $\rho_{\text{coh}}(N) = \sigma(N) \cdot R_{\text{surf}} \cdot m_p / (V_C \cdot R_{\text{tot}})$ unconditional. $C_{\mu\nu}^{(\text{mass})} = \rho_{\text{coh}} c^2 u_\mu u_\nu$ unconditional; $C^{(\text{spin})}$ and $C^{(\text{orb})}$ theorems; $C^{(\text{leak})}$ fully resolved via D51+D52. PDL Einstein equation with explicit source (Corollary 9.1).
D49	Derivation of the London Equation from the PDL Axioms: Resolution of OP-London [56]	Theorem OP-London resolved. Lemma: $\nu = \frac{1}{4} \ \psi_1 - T^k \psi_1\ _{\mathcal{H}_{\text{cyc}}}^2$ verified 768/768 in exact integer arithmetic. Axiom C4 forces $\psi_j = \psi_1$ over V_C (via C3 connectivity), hence $\phi = \text{const}$ (London gauge). Corollary: $j_s^i = -(n_{\text{coh}} e^2 / m_p c) A^i$ with $n_{\text{coh}} = \sigma(N) R_{\text{surf}} / (V_C R_{\text{tot}}) \in \mathbb{Q}(\sqrt{5})$; $\lambda_{L, \text{PDL}}(N = 40) \approx 7.25 \times 10^{-15} \text{ m}$.

Label	Title (abbreviated)	Main result / role
D50	Derivation of the Bekenstein–Hawking One-Quarter Coefficient from the PDL Axioms: Resolution of OP12/BH-3 [57]	Theorem OP12/BH-3 resolved. The coefficient $\frac{1}{4}$ in $S_{\text{BH}} = k_B A / (4\ell_P^2)$ equals the stable fraction $4/16 = 1/4$ per surface relation, proved from D29 (exhaustive, 768/768) and H3 (D42). Statistical independence $P(e_1 \wedge e_2) = 1/16 = (1/4)^2$ verified over 30 720 cases. $\Omega_{\text{surf}} = 4^{R_{\text{surf}}}$; $S_{\text{surf}} = k_B R_{\text{surf}} \ln 4 = 695.352$ nats. Black hole thermodynamics layer complete.
D51	The Cosmological Leakage Constant from the PDL Axioms: A Structural Derivation of C and Resolution of OP1-D35 [58]	Theorem OP1-D35 structurally resolved. $\beta_1(K_4) = 3$ gives three leakage cycles (topology). C1+C3 forces prime cycle lengths. Indices $k_1 = n_d - n_u + R_e - 1 = 9$, $k_2 = n_u - R_e + 1 = 19$, $k_3 = R_e \cdot n_d = 168$ are exact theorems; $k_1 + k_2 = n_d$. Conjecture PDL-C: $C = (1 - \kappa)^{997} \times (930/11017)^{23} \times (1 - \eta_L)^{67}$ at 0.17 ppm of Λ_{obs} . Uniqueness verified by exhaustive exclusion of all alternative bases.
D52	Formal Identification of the Three Leakage Bases as Natural Representatives of the Three Cycles of $\beta_1(K_4) = 3$: Completion of Conjecture PDL-C [59]	Theorem OP1-D35 fully resolved. The three bases $(1 - \kappa)$, $R_{\text{val}}/R_{\text{tot}}$, $(1 - \eta_L)$ are proved to be the natural and unique representatives of the three leakage cycles: $(1 - \kappa)$ from D42 (surface sector), $R_{\text{val}}/R_{\text{tot}}$ from D29 Lemma 2 ($\Omega_{\text{val}} = 1$, valence sector), $(1 - \eta_L)$ from D17/D30 (coupling sector). Algebraic independence verified (40 401 cases). Full causal chain: $C1\text{--}C4 \rightarrow K_4 \rightarrow \beta_1 = 3 \rightarrow (p_{k_1}, p_{k_2}, p_{k_3}) \rightarrow C \rightarrow \Lambda_{\text{PDL}}$. Cosmological leakage layer complete.
D53	From Four Axioms to the Cosmological Constant: Causal Closure of the Chain C1–C4 to Λ — A Synthesis and Computational Verification [60]	Synthesis and verification of the complete chain $C1\text{--}C4 \rightarrow \Lambda_{\text{PDL}}$ in a single self-contained document. Independent numerical verification at 50-decimal precision: C_{PDL} confirmed to 0.41 ppm relative to D51; prime robustness confirmed (composite substitution degrades agreement by ≥ 4 orders of magnitude); metrological origin of the apparent 8.17 ppm offset identified (8.6 ppm in $(m_p c / \hbar)^2$ between CODATA versions, no structural significance). The chain spans 57 orders of magnitude.

2.2 Reading guide for a new collaborator

The following reading pathways are recommended depending on one’s entry point.

- **Understanding the foundations.** Start with D02 (conceptual overview), then D16a (proof that K_4 is the unique minimal closure), then D01 (full axiomatic development). For the proton architecture, read D16 and D16b for the combinatorial selection arguments.
- **Understanding the constants.** Read D12 for α , then D21 and D25 for the α – G bridge. Read D23 for the topological proof that the exponent 18 is necessary. Then read D29 and

D30 for the Gate 1 and Gate 2 proofs. Read D36 for the final Gate 3 proof, and D39/D42 for the full resolution of OP1.

- **Understanding the quantum dynamics.** Read D32 (Schrödinger), D33 (Dirac), D34 (Born Level 1), D35 (Einstein), and D46 (Born Level 2, $U(1)$) in order.
- **Understanding the cosmology.** Read D24, D27, and D35.
- **Understanding black hole thermodynamics.** Read D37 (area law), then D38 (Bekenstein–Hawking and PBH predictions), then D45 (falsifiable confrontation with Fermi-LAT).
- **Understanding nuclear stability and the periodic table.** Read D22, D23, D40, D41, and D47.
- **Understanding the London equation and superconductivity analogy.** Read D46 ($U(1)$ phase freedom), then D49 (London equation from C4 and the metric identity $\nu = \frac{1}{4}\|\psi_1 - T^k\psi_1\|^2$). D50 completes the black hole thermodynamics layer with the one-quarter coefficient.
- **Understanding the cosmological constant.** Read D48 for the leakage term structure, D51 for the derivation of Conjecture PDL-C and the role of prime cycle lengths from $\beta_1(K_4) = 3$, and D52 for the formal identification of the three leakage bases. For a self-contained synthesis of the complete chain with numerical verification, read D53.
- **The closure of the foundational programme.** Read D39 first for context, then D42 for the full proof of H3. After D42, all results previously conditional on H3 are unconditional theorems. Then read D43 (geometric leakage, OP-A) and D44 (filter factor k , OP-B) for the complete causal chain to G . Finally, D51+D52 close the chain to Λ_{PDL} .
- **The coherence stress-energy tensor.** Read D35 for the structural form, then D48 for the explicit component derivation.
- **Numerical verification and the computational infrastructure.** Read D43. It provides self-contained Colab code for every major result and defines the verification protocol for future documents.
- **Continuing the programme.** Read Section 13 for the complete prioritised list of open problems, and Section 14 for the critical-path dependency map.

3 Axiomatic Foundations

3.1 The four relational axioms C1–C4

A logical configuration in PDL is a finite signed graph $G = (V, E, s)$, where V is a finite set of vertices (informational entities), E is a set of undirected edges (binary relations), and $s: E \rightarrow \{1, -1\}$ assigns a sign (agreement or disagreement) to each relation [1, 2]. Four relational axioms are imposed.

Axiom 3.1 (C1 — Minimal binary pulsation). Existence is instantiated as a repeatable, non-trivial alternation between two discrete states, independent of any pre-existing temporal metric. Formally, the discrete dynamical system defined on the space of sign configurations must admit at least one logical 2-cycle: a configuration that returns to itself after exactly two update steps and is not a fixed point.

Axiom 3.2 (C2 — Coherence under logical stress). Every admissible configuration must maintain internal consistency under relational stress. Formally, the graph must be *balanced* in the sense of Harary: every cycle of relations carries a positive product of signs. This is equivalent to requiring that the graph be 2-colourable with respect to signed adjacency.

Axiom 3.3 (C3 — Completeness of relational coverage). Every pair of informational entities must be related, directly or indirectly, by at least one signed relation. Formally, the signed graph G must be connected.

Axiom 3.4 (C4 — Optimisation of coherence cost). Among all configurations satisfying C1–C3, the physically realised configuration minimises the total coherence cost, defined as the fraction of violated triangles (triangles with negative triple product). This is a discrete analogue of a variational principle.

3.2 The minimal admissible closure: K_4

Theorem 3.5 ([19]). *The complete signed graph K_4 on four vertices and six edges is the unique minimal admissible closure of axioms C1–C4: no admissible closure exists on $|V| < 3$ vertices, and K_4 is the first connected balanced complete graph admitting a non-trivial 2-cycle. It is identified with the electron prototype, with relational budget $R_e = 6$.*

3.3 The proton quintuplet

The proton is the minimal hierarchical composite closure admissible under C1–C4, characterised by the integer quintuplet $(n_u, n_d, \sigma, R_{\text{sea}}, R_{\text{tot}}) = (24, 28, 930, 10087, 11017)$, where $\nu_u = 24$ is the number of u -valence cores, $\nu_d = 28$ the number of d -valence cores, $\sigma = 930$ the active surface, $R_{\text{sea}} = 10087$ the sea relational budget, and $R_{\text{tot}} = 11017$ the total relational content [1, 18–20]. The neutron quintuplet is $(24, 28, 1032, 9960, 10992)$ [28].

Conjecture 3.6 (Global uniqueness [18, 20]). The quintuplet $(24, 28, 930, 10087, 11017)$ is the *global* (not merely local) minimum of the coherence leakage functional over all admissible integer tuples. The local uniqueness has been verified computationally for a neighbourhood of radius 10^4 (D43). The global statement remains open (OP2).

3.4 The primary leakage parameter κ

Under the Indifference Lemma H3 (now a theorem, D42) and D39, the primary leakage parameter is:

$$\kappa = \frac{310\varphi}{11017} \in \mathbb{Q}(\sqrt{5}), \quad \varphi = \frac{1 + \sqrt{5}}{2}, \quad \kappa \approx 0.0075194. \quad (1)$$

This is an unconditional theorem of C1–C4 [46, 49]. Under H3 (now a theorem), H1 follows as a theorem [46]: the two-factor decomposition $\kappa = 3/\sigma$ holds. H3 has been derived from C1–C4 in D42 via three lemmas D42-L1, D42-L2, D42-L3; it is no longer a named hypothesis but an unconditional theorem of the programme.

Theorem 3.7 (Gate 3, [43]). *Since H3 is now a theorem [49], H1 follows unconditionally [46], and Gate 3 holds without any named hypothesis: $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ and $\sigma_0 \equiv 0$. The independence of gravitational engagements ($\sigma_0 = 0$ exactly) follows from the AB coupling criterion without additional hypothesis. The tracelessness $\text{Tr}(A) = 0$ of the generator matrix follows from the combinatorial structure of K_4 alone.*

3.5 The proton–electron mass ratio μ

The PDL prediction for the proton–electron mass ratio is $\mu \approx 1836.152\,670$, a parameter-free result consistent with the experimental value $1836.152\,673\,43(11)$ at the level of 1.8×10^{-9} [35–37]. This is the programme’s primary near-term falsifiable prediction; current precision measurements can test it at 0.001 ppm.

4 The Three Gates: Critical Proofs

4.1 Gate 1: Axiomatic derivation of 155/11017

Theorem 4.1 (Gate 1, [36]). *The amplitude 155/11017 appearing in the mass-ratio correction formula is derivable from PDL axioms C1–C4 alone, via the AB coupling criterion. Specifically: the stability criterion AB has been verified exhaustively over all 768 admissible configurations; in every stable case, $P_2/P_1 = -1$ exactly; and the combinatorial count yields 155/11017 without any free parameter.*

4.2 Gate 2: The QCD correction coefficient $a = 2$

Theorem 4.2 (Gate 2, [37]). *The QCD correction coefficient $a = 2$ in the mass-ratio formula $\mu = \frac{155}{11017}(1 - 2\kappa^2)\frac{m_{\text{iso}}}{m_p}$ is structurally forced by axiom C1 (minimal binary pulsation): the factor 2 counts the two distinct pulsation branches of the proton’s valence core. Furthermore, $m_{\text{iso}} = m_d - m_u$ is the unique irreducible external parameter at the PDL–QCD interface; it cannot be derived from C1–C4 alone.*

4.3 Gate 3: $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$

Gate 3 was first established in D31 under a linear bridge inversion hypothesis. D36 strengthened this proof by: proving that the independence of gravitational engagements ($\sigma_0 = 0$) follows exactly from AB, without any additional assumption; proving that $\text{Tr}(A) = 0$ from the combinatorial structure of K_4 alone; and reducing the remaining hypothesis to H1 only (H2 was absorbed into the proof). D39 further reduced H1 to H3: given the Indifference Lemma, H1 follows as a theorem. D42 then proved H3 from C1–C4 unconditionally. See Theorem 3.7 for the full statement.

5 Derived Dynamical Equations

5.1 The Schrödinger equation (D32)

Theorem 5.1 ([39]). *In the continuum limit of the PDL coherence-cycle dynamics, the AB coupling criterion forces the amplitude evolution to satisfy the Schrödinger equation $i\hbar \partial_t \psi = -\frac{\hbar^2}{2m_e} \nabla^2 \psi + V\psi$. The diffusion parameter $b = -i\hbar/(2m_e)\partial_x^2$ is uniquely forced by the AB stability requirement; no free parameter enters.*

5.2 The Dirac equation (D33)

Theorem 5.2 ([40]). *Among the eight candidates for the time-reversal operator T acting on the $\text{SL}(2, \mathbb{C})$ pulsation of K_4 , the unique admissible choice consistent with AB is $T = -i\sigma_2$ (where σ_2 is the second Pauli matrix). This forces $T^2 = -I_2$, which implies period-4 dynamics and hence spin- $\frac{1}{2}$ statistics. The Clifford algebra of Dirac matrices then follows as a theorem (residual $< 0.001\%$), and the non-relativistic limit reproduces the result of Theorem 5.1 exactly.*

5.3 Born's rule (D34 and D46)

Theorem 5.3 (Level 1, [41]). *Let $\sigma \in \{1, -1\}$ be the combinatorial signature of a coherence cycle in the PDL framework. Under the apparatus symmetry assumption, the leakage probability satisfies $p_{\text{coh}} \propto \sigma^2$. The five Gleason axioms (countability, null-assignment, covariance, normalisation, and regularity) are verified for this measure. By the Gleason uniqueness theorem for spin- $\frac{1}{2}$ systems, the probability assignment is uniquely given by Born's rule $P(\psi) = |\langle \psi | \phi \rangle|^2$.*

Theorem 5.4 (Level 2, OP4 resolved, [53]). *The $U(1)$ phase freedom $\psi \mapsto e^{i\alpha}\psi$ of quantum mechanics is a structural consequence of the K_4 pulsation, not an independent postulate. Specifically: (Proposition 1) the global sign equivalence $s \sim -s$ of K_4 coherent configurations, combined with $T^2 = -I_2$ (D33), generates a canonical $U(1)$ action on the amplitude space $\mathcal{H}_{\text{cyc}} \cong \mathbb{C}^2$, whose orbits are the fibres of the Hopf fibration $S^1 \hookrightarrow S^3 \rightarrow S^2$; (Proposition 2) the $(A) \wedge (B)$ stability criterion is $U(1)$ -invariant, verified algebraically over all 8 coherent configurations; (Proposition 3) the unique $U(1)$ -equivariant probability measure on S^3 consistent with the Level 1 Gleason axioms is Born's rule $P = |\langle \theta | \psi \rangle|^2$ on the base $\mathbb{P}^1(\mathbb{C})$. This completes the quantum dynamics layer of the PDL programme.*

5.4 The Einstein coupling equation (D35 and D48)

Theorem 5.5 ([42, 55]). *Under Gate 3 (now unconditional via D42), the PDL gravitational coupling satisfies $G_{\text{eff}}(N) = G_{\text{Newton}}$, and the PDL Einstein field equation takes the form $G_{\mu\nu}^{\text{eff}} = 8\pi G(N) C_{\text{coh}}$, where $C_{\text{coh}} = C^{(\text{mass})} + C^{(\text{spin})} + C^{(\text{orb})} + C^{(\text{leak})}$ is the coherence stress-energy tensor with explicit components derived in D48. The Hubble tension ratio $H_{0,\text{local}}/H_{0,\text{CMB}} = 1.085868$ versus the PDL prediction 1.085935 gives a 0.006% relative error with zero free parameters.*

6 Black Hole Thermodynamics

6.1 The area law as an unconditional theorem (D37)

Theorem 6.1 (BH-1, [44]). *The information-theoretic entropy of a PDL closure is carried entirely by its active surface R_{surf} , not by its total relational budget R_{tot} . Formally: (1) the valence sector has $S_{\text{val}} = k_B \ln 1 = 0$ [36]: the AB-stable valence configuration is unique; (2) the sea sector is excluded from stationary coupling [36], hence $S_{\text{sea}} = 0$; (3) the total entropy is $S = k_B \ln R_{\text{surf}}$. This is an unconditional theorem; it requires no external geometric assumption and no hypothesis beyond C1–C4 and D29.*

Corollary 6.2 (PDL area law, [44]). *Under Proposition 6.1 and the identification of R_{surf} with the physical horizon area via $G_{\text{eff}}(N)$ (Gate 3, Theorem 3.7), the Bekenstein–Hawking area law $S \propto A$ is a structural consequence of the four PDL axioms. It is not an independent postulate.*

6.2 The Bekenstein–Hawking formula (D38)

Theorem 6.3 (BH-2, [45]). *Under Gate 3 ($G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$, now unconditional), substituting the PDL effective gravitational coupling into the Bekenstein–Hawking formula $S_{\text{BH}} = k_B c^3 A / (4\hbar G(N))$ yields the algebraic identity $S_{\text{BH}}/k_B = 4\pi (M_{\text{eff}}/M_{\text{Pl}})^2$, where $M_{\text{Pl}} = (\hbar c/G_{\text{PDL}})^{1/2}$. This identity is verified numerically to relative error $< 10^{-15}$ over ten decades of black hole mass. The coefficient 4π is not a free parameter; it is fixed by the Schwarzschild geometry.*

Three falsifiable, parameter-free predictions arise from Theorem 6.3 [45]: (1) CMB-epoch entropy suppression by a factor ≈ 0.848 at $N_{\text{CMB}} = 40$; (2) primordial black hole survival threshold shifted upward by 11.89% relative to GR, testable with Fermi-LAT via BLACKHAWK/ISATIS [52]; (3) redshift-dependent S/E ratio in AGN populations traceable with DESI/Euclid.

7 The Coherence Stress-Energy Tensor (D48)

7.1 Double determination of the coherence volume

Theorem 7.1 (Double determination of V_C , [55]). *The coherence volume $V_C = (4/3)\pi[\hbar/(m_p c)]^3$ is forced by two independent chains from C1–C4, without free parameter: (Chain 1, temporal, D10a) the pulsation frequency $\omega_p = m_p c^2/\hbar$ forces the Compton wavelength $\lambda_C = \hbar/(m_p c)$ and hence $V_C = (4/3)\pi\lambda_C^3$; (Chain 2, geometric, D23) the S^2 topology forces $n = 3$ and the $(4/3)\pi$ factor, while $R_{\text{geom}} = \lambda_C$ is algebraically exact. Numerical agreement: $V_{\text{geom}}/V_C = 1.00000000$ at machine precision (float64).*

7.2 The coherence mass density

Theorem 7.2 ([55]). *The coherence mass density of an ensemble of N nucleons is*

$$\rho_{\text{coh}}(N) = \sigma(N) \cdot \frac{R_{\text{surf}} \cdot m_p}{V_C \cdot R_{\text{tot}}}, \quad (2)$$

where $\sigma(N) = 1 - (1 - \kappa)^N \in \mathbb{Q}(\sqrt{5})$. This is an unconditional theorem of C1–C4.

7.3 Components of the coherence stress-energy tensor

Theorem 7.3 (Explicit form of C_{coh} , [55]). *The coherence stress-energy tensor appearing in the PDL Einstein equation decomposes as $C_{\text{coh}} = C^{(\text{mass})} + C^{(\text{spin})} + C^{(\text{orb})} + C^{(\text{leak})}$, with the following status for each component:*

1. $C_{\mu\nu}^{(\text{mass})} = \rho_{\text{coh}}(N) c^2 u_\mu u_\nu$ (*unconditional theorem*).
2. $C_{\mu\nu}^{(\text{spin})} \sim \hbar^2 \rho_{\text{coh}}^2 / (m_p c^2)$ (*theorem; correction $\sim 10^{-12}$ relative to $C^{(\text{mass})}$*).
3. $C_{\mu\nu}^{(\text{orb})} = m_p \langle j_\mu j_\nu \rangle / \rho_{\text{coh}}$ (*theorem, state-dependent via D32; connection to London equation via D46 conjectured as OP-London*).
4. $C_{\mu\nu}^{(\text{leak})} = -C \eta_L^{18} (m_p c / \hbar)^2 g_{\mu\nu}$ (η_L^{18} is a *theorem* from D17 and D30; the prefactor C is *open* as OP1–D35, bounded observationally at $C \approx 8.16 \times 10^{-46}$ from Planck 2020).

Corollary 7.4 (PDL Einstein equation with explicit source, [55]). *The PDL Einstein field equation with complete explicit source is:*

$$G_{\mu\nu}^{\text{eff}} = 8\pi G(N) \left[\rho_{\text{coh}} c^2 u_\mu u_\nu + C_{\mu\nu}^{(\text{spin})} + C_{\mu\nu}^{(\text{orb})} - C \eta_L^{18} (m_p c / \hbar)^2 g_{\mu\nu} \right]. \quad (3)$$

The constant C is bounded observationally but remains to be derived from C1–C4 (OP1–D35).

8 Nuclear Stability and the Periodic Table

8.1 The neutron survival condition (D40)

Theorem 8.1 (Regime I, [47]). *The identity $T \equiv (\Delta n + 1)^2 = 1.010414 \approx 1$ is a consequence of the proton quintuplet. The neutron survival condition $N_{\min}(T, Z) = \lceil n_1/2 \rceil$ therefore entails $N_{\min}(Z) = Z$ for all $Z \leq 20$, exactly. This is an unconditional theorem; no hypothesis beyond the quintuplet values is required.*

8.2 Conflict saturation (D40)

Theorem 8.2 (Saturation, [47]). *For all $Z > 20$, the total proton–proton conflict saturates: $C(Z > 20) = 190 T_{pp}$ exactly. Every nucleus with $Z > 20$ presents the same total conflict to its neutron sea, independently of the number of protons. The observed increase of $N(Z)$ beyond $Z = 20$ is driven not by increasing conflict but by the structural requirement that each neutron maintains its own coherence.*

8.3 Spin–orbit splitting from axioms (D47)

Theorem 8.3 (OP13 resolved, [54]). *The value $\Delta n = 4$ is the unique element of $\{0, 4, 8, 12, \dots, 32\}$ such that the discriminant of the quasi-completeness equation $3n_u^2 + (2\Delta n - 3)n_u + \Delta n(\Delta n - 1) - 1860 = 0$ is a perfect square. The discriminant equals $149^2 = 22201$ for $\Delta n = 4$ and fails to be a perfect square for all other admissible values (exhaustive verification). This forces $n_u = 24$, $n_d = 28$, and $s = \Delta n/(2n_u) = 1/12$ as unconditional theorems of C1–C4.*

Lemma 8.4 (Mirror Lemma, [54]). *The harmonic oscillator PDL levels of the proton and neutron are isomorphic: $n_u(n) = n_u(p) = 24$ and $n_d(n) = n_d(p) = 28$. Proton and neutron share the same splitting $1/12$, the same level order, and the same degeneracies. Verified analytically for the first 20 levels; zero exception.*

8.4 Sub-shell filling rates and the periodic table (D47)

Theorem 8.5 (OP14 resolved, [54]). *The sub-shell filling rates for all $Z = 1–82$ are:*

$$r_{\text{exc}}(Z) = \begin{cases} 0 & \text{if } Z \in \{28, 50, 82\} \text{ (magic closures, from OP13),} \\ 1 & \text{otherwise (mirror filling, ratio } 2/2 = 1 \text{ independent of degeneracy } d). \end{cases} \quad (4)$$

Verified exhaustively for 31 sub-shell boundaries (31/31), zero free parameter, zero external input.

Corollary 8.6 (Valley of stability as an unconditional theorem, [54]). *The formula $N_{\min}(Z) = Z$ for $Z \leq 20$ and $N_{\min}(Z) = \lfloor 20 + r_{\text{exc}}(Z) \cdot (Z - 20) \rfloor$ for $Z > 20$ is an unconditional theorem of C1–C4, with $r_{\text{exc}}(Z)$ given by Theorem 8.5. The periodic table (valley of stability for $Z = 1–82$) is a corollary of the four PDL axioms.*

8.5 Nuclear spectroscopy confrontation (D41)

Document D41 provides the first direct confrontation of the PDL nuclear framework with a published nuclear spectroscopy experiment. The experiment of Ha et al. [61] measures the first-excited-state lifetimes in ^{84}Mo and ^{86}Mo . Three independent observables are consistent with the PDL structural identity $T = (\Delta n + 1)^2 = 25$: (1) the BE2 ratio (22% discrepancy, within the 33% experimental uncertainty on ^{84}Mo); (2) the wave-function component ratio $104/53 = 1.962$ within 2% of the PDL prediction 2.000; (3) the Shannon entropy difference (25%). Two falsifiable predictions for $^{88}\text{Ru}/^{90}\text{Ru}$ and $^{92}\text{Pd}/^{94}\text{Pd}$ follow from Conjecture H_B (OP17).

Conjecture 8.7 (H_B — Closure multiplicity, [48]). *For a nucleus at the boundary of an island of inversion dominated by a kp - kh intruder configuration, the number of accessible partial closures scales as $N_{\text{comp}}(k) = k!$, yielding $N_{\text{comp}}(k_1)/N_{\text{comp}}(k_2) = k_1!/k_2!$. For $k_1 = 8$ (^{84}Mo) and $k_2 = 4$ (^{86}Mo), the predicted ratio is 2.000 exactly.*

9 Cosmology

9.1 The density function $\sigma(N)(N)$

For an ensemble of N protons (or nucleons), the effective coherence engagement fraction is $\sigma(N) = 1 - (1 - \kappa)^N \in \mathbb{Q}(\sqrt{5})$, where κ is the surface-to-total ratio of the proton quintuplet [31, 33]. As $N \rightarrow \infty$, $\sigma(N) \rightarrow 1$, recovering κ_G in the dense limit. As $N \rightarrow 0$, $\sigma(N) \approx N\kappa$, giving a suppressed coupling at low densities.

9.2 The Hubble tension resolution

Theorem 9.1 ([34, 42]). *Let $n_6 = 6n_-/n_+$, where n_- is the neutron magnetic moment in nuclear magnetons and $n_+ = 10992$ is the total relational content of the neutron. Then $n_6 \approx 40.102$, so $N_{\text{CMB}} = \lfloor n_6 \rfloor = 40$. The CMB-epoch Hubble constant is $H_{0,\text{CMB}} = 67.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$, which agrees with the Planck 2018 measurement to within 0.27σ . The ratio $H_{0,\text{local}}/H_{0,\text{CMB}} = \sigma(N)(N_{\text{local}})/\sigma(N)(N_{\text{CMB}})$ reproduces the observed Hubble tension at 0.006% with zero free parameters.*

10 The PDL–QCD Interface

The unique irreducible external parameter of the programme is $m_{\text{iso}} = m_d - m_u$, the up–down quark mass difference. This quantity arises at the PDL–QCD boundary (Theorem 4.2) and cannot be derived from the relational axioms alone.

Current value and uncertainty. The PDL structural derivation yields $m_{\text{iso}} \approx 2.532 \text{ MeV}$, to be compared with the FLAG lattice QCD result $m_{\text{iso}} = 2.529 \pm 0.030 \text{ MeV}$ [37]. This agreement at the sub-percent level is non-trivial, since PDL derives this value from the quintuplet and the observed μ , with no lattice QCD input.

Residual. A 47 ppm residual remains in the mass-ratio formula; the structural factor g accounting for it has not yet been identified (Open Problem 13.8).

11 Causal Chain Closure and Geometric Leakage (D43–D44)

Documents D43 [50] and D44 [51] together close the causal chain from axioms to Newton’s constant.

D43 — Geometric leakage and OP-A resolution. D43 establishes the computational backbone of the programme (Python/SymPy at 50-decimal precision) and proves the geometric leakage parameter as an unconditional theorem. The general analytic formula for the boundary energy is:

$$E_{\text{bord}} = A \cdot n_{u\text{-cores}} + B \cdot n_{d\text{-cores}} + (\Delta n + 1)^2, \quad A = 55, \quad B = 194, \quad (5)$$

where $c = 3$ is forced by the K_4 3-regularity. This yields $\varepsilon_{\text{geom}}(p) = 329/10087$ and $\varepsilon_{\text{geom}}(n) = 468/9960$ as unconditional theorems of C1–C4 (OP-A resolved).

D44 — Filter factor k and OP-B resolution. D44 derives $k = 0.921716$ from C1–C4 via the trace structure of K_4 and the S_4 -equivariance constraints proved in D42. The complete causal chain is:

$$\text{C1–C4} \rightarrow K_4 \rightarrow \text{quintuplet} \rightarrow \varepsilon_{\text{geom}} \rightarrow \varepsilon_G = \varepsilon_{\text{geom}} \times k^{18} \rightarrow G.$$

No remaining open problem exists in this chain. As a corollary, OP8 is resolved: the 17 ppm residual of D28 is attributed to the CODATA measurement precision of G , not to any structural gap.

12 Epistemic Architecture

Table 2 Nine-layer epistemic architecture of the PDL programme (updated for v20). Each layer lists the status of its key claims and the primary documents supporting them.

Layer	Domain	Key claims and status	Primary documents
L0	Computational verification	Infrastructure: 18 filters at 50-decimal precision (D43). $\varepsilon_{\text{geom}}(p) = 329/10087$ and $\varepsilon_{\text{geom}}(n) = 468/9960$ proved (OP-A). $k = 0.921716$ proved (OP-B, D44). ε_G^B closed-form in $\mathbb{Q}(\sqrt{5})$ exact (OP8 resolved).	D43, D44
L1	Axioms / combinatorial results	Theorems: K_4 unique [19]; $\nu_u = 24$ unique (D47 OP13); $18 = 6 + 5 + 4 + 3$ [30]; $\sigma = 930$ unique; $a = 2$ [37]; $\sigma(N) = 1 - (1 - \kappa)^N$ [31]; AB criterion [36]; $T = -i\sigma_2$ unique [40]; Clifford algebra [40]; period-4 [40]; $\sigma_0 \equiv 0$ [43]; $\text{Tr}(A) = 0$ [43]; $S_{\text{val}} = 1$ [44]; $\Delta n = 4$ unique (D47). No free parameter; no external input.	D01, D16a, D23, D29, D30, D33, D36, D37, D47
L2	QCD interface	Minimal, forced: $m_{\text{iso}} = m_d - m_u$ is the unique irreducible external input, forced by C1, not chosen. Conjecture: $m_{\text{iso}} \approx 2.532$ MeV.	D28, D30
L3	Cosmological coupling	Theorem: $\kappa = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$ unconditional (D42). $G_{\text{eff}}(N) = \sigma(N) \cdot G_{\text{PDL}}$ unconditional.	D24, D31, D36, D39, D42
L4	Quantum dynamics [COM- PLETE]	Theorems: Schrödinger [39]; Dirac [40]; Born Level 1 [41]; Born Level 2 + U(1) (OP4 resolved) [53]; London equation (OP-London resolved) [56].	D32, D33, D34, D46, D49

Layer	Domain	Key claims and status	Primary documents
L5	Gravitational and black hole dynamics [COM- PLETE]	Theorem (unconditional): $G_{\mu\nu}^{\text{eff}} = 8\pi G(N)C_{\text{coh}}$. Theorem : area law $S \propto R_{\text{surf}}$ unconditional [44]. Theorem : $S_{\text{BH}}/k_B = 4\pi(M_{\text{eff}}/M_{\text{Pl}})^2$ [45]. Theorem : coefficient $\frac{1}{4}$ (OP12 resolved) [57]. C_{coh} explicit: all four components theorems [55, 58, 59].	D35, D36, D37, D38, D48, D49, D50
L6	Nuclear stability / periodic table [COMPLETE]	Theorems : $N_{\min}(Z \leq 20) = Z$ exact; $C(Z > 20) = 190 T_{pp}$ [47]; $\Delta n = 4$ unique (OP13) [54]; Mirror Lemma [54]; $r_{\text{exc}}(Z)$ 31/31 (OP14) [54]; $N_{\min}(Z)$ $Z = 1\text{--}82$ unconditional corollary [54].	D22, D23, D40, D47
L7	Falsifiable predictions	$\mu \approx 1836.152\,670$ testable at 0.001 ppm; $m_{\text{iso}} \approx 2.532$ MeV vs. FLAG; PBH threshold $M_{\text{PDL}}^* = 5.706 \times 10^{14}$ g (+11.89%, D45); spectral cutoff at 93–104 MeV in IGRB (D45); $H_{0,\text{CMB}} = 67.26$ km s $^{-1}$ Mpc $^{-1}$; AGN $S/E \propto \sigma(N(z))$; $B(E2)$ ratio and N_{comp} ratio in $^{88}\text{Ru}/^{92}\text{Pd}$; $\lambda_{L,\text{PDL}}(N = 40) \approx 7.25 \times 10^{-15}$ m (D49).	D28–D30, D35, D38, D41, D45, D49
L8	Cosmological leakage [COM- PLETE]	Theorems : $\beta_1(K_4) = 3$ (topology, D23); C1+C3 forces prime exponents (D51); indices $k_1 = 9$, $k_2 = 19$, $k_3 = 168$ exact (D51); bases $(1 - \kappa)$, $R_{\text{val}}/R_{\text{tot}}$, $(1 - \eta_L)$ uniquely identified (D52). $C = (1 - \kappa)^{997} \times (930/11017)^{23} \times (1 - \eta_L)^{67}$ at 0.17 ppm; $\Lambda_{\text{PDL}} \approx 1.089 \times 10^{-52}$ m $^{-2}$. Full chain C1–C4 $\rightarrow \Lambda_{\text{PDL}}$ complete. Independent numerical verification 0.41 ppm (D53).	D17, D23, D29, D30, D42, D51, D52, D53

13 Open Problems

The following list supersedes the open-problem list of v19. Problems are classified by priority and domain. OP1, OP4, OP8, OP11, OP12, OP13, OP14, OP-London, and OP1-D35 have been fully resolved and are retained for historical reference only.

Open Problem 13.1 (OP1 — Resolved: Indifference Lemma H3). [Fully resolved by D42](#). The Indifference Lemma (H3) has been derived from axioms C1–C4 via three lemmas: D42-L1 (Equiparticipation), D42-L2 (Cross-Triangle Blindness), D42-L3 (S_4 -Equivariance). As a corollary, $\kappa = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$ is an unconditional theorem. The entire causal chain from

C1–C4 to the Bekenstein–Hawking entropy is now fully axiomatic.

Open Problem 13.2 (OP1–D35 — Resolved: cosmological leakage constant C). [Fully resolved by D51+D52](#). The constant C in $C_{\mu\nu}^{(\text{leak})} = -C \eta_L^{18} (m_p c / \hbar)^2 g_{\mu\nu}$ is $C = (1 - \kappa)^{997} \times (930/11017)^{23} \times (1 - \eta_L)^{67}$, derived from C1–C4 via $\beta_1(K_4) = 3$ (D23), prime cycle lengths (D51 Lemma 2), quintuplet indices (D51 Lemma 3), and identification of the three leakage bases (D52). Agreement with Λ_{obs} : 0.17 ppm. The causal chain C1–C4 $\rightarrow \Lambda_{\text{PDL}}$ is complete.

Open Problem 13.3 (OP2 — Global uniqueness of the proton quintuplet). Prove that (24, 28, 930, 10087, 11017) is the *global* (not merely local) minimum of the coherence leakage functional over all admissible integer tuples. The local uniqueness conjecture has been verified computationally for all tuples within a neighbourhood of radius 10^4 ; the global statement remains open. **Priority:** high. **Entry point:** D16, D16b.

Open Problem 13.4 (OP3 — Algebraic proof of 155/11017). Provide a closed-form algebraic derivation of 155/11017 from the combinatorial structure of K_4 , replacing the exhaustive enumeration over 768 configurations in D29. The exhaustive verification is complete and rigorous, but an analytic proof would yield structural insight. **Priority:** medium. **Entry point:** D29.

Open Problem 13.5 (OP4 — Resolved: Born’s rule Level 2). [Fully resolved by D46](#). The U(1) phase freedom $\psi \mapsto e^{i\alpha}\psi$ has been derived from the K_4 pulsation structure via three propositions establishing the Hopf fibration structure, U(1)-invariance of the AB criterion, and uniqueness of the Born measure. The quantum dynamics layer is complete.

Open Problem 13.6 (OP5 — CMB angular power spectrum from PDL). Derive the CMB angular power spectrum C_ℓ from the PDL cosmological framework, using the $\sigma(N)$ -modified gravitational coupling. The scalar spectral index $n_s \approx 0.965$ and the tensor-to-scalar ratio r are the primary targets. **Priority:** medium. **Entry point:** D27, D35.

Open Problem 13.7 (OP6 — PDL prediction for gravitational wave polarisation). Determine whether the PDL Einstein equation (with the $\sigma(N)$ -modified coupling) predicts any deviation from GR in the polarisation or amplitude of gravitational waves at cosmological scales. **Priority:** medium. **Entry point:** D35, D38, D48.

Open Problem 13.8 (OP7 — Derivation of the 47 ppm residual in μ). Identify the combinatorial factor g responsible for the 47 ppm residual in the mass-ratio formula and derive it from C1–C4. Gate 1 (D29) and Gate 2 (D30) account for the leading structure. **Priority:** high. **Entry point:** D28, D29, D30, D43.

Open Problem 13.9 (OP8 — Resolved: closed-form for ε_G^B in $\mathbb{Q}(\sqrt{5})$). [Resolved by D44](#). The formula $\varepsilon_G^B = \kappa \cdot C / (6 + \kappa) \cdot (1 + 2/R_{\text{tot}}) \in \mathbb{Q}(\sqrt{5})$ is an exact closed-form expression. The 17 ppm residual relative to CODATA reflects the measurement precision of G , not a structural gap.

Open Problem 13.10 (OP9 — Extension to higher generations: muon and tau). Generalise the PDL proton architecture to the second and third generations of the Standard Model. The muon–electron and tau–electron mass ratios should be derivable from hierarchical extensions of the quintuplet. **Priority:** medium. **Entry point:** D01, D12.

Open Problem 13.11 (OP10 — Weak force and W/Z mass ratios). Derive the Weinberg angle $\sin^2 \theta_W \approx 0.231$ and the W/Z mass ratio from the PDL coherence structure. **Priority:** medium. **Entry point:** D33, D34, D46.

Open Problem 13.12 (OP11 — Resolved: dark energy as cosmological leakage). [Fully resolved by D51+D52 \(synthesis D53\)](#). The cosmological constant $\Lambda \approx 1.089 \times 10^{-52} \text{ m}^{-2}$ is derived from C1–C4 as $\Lambda_{\text{PDL}} = C \cdot \eta_L^{18} \cdot (m_p c / \hbar)^2$ where $C = (1 - \kappa)^{997} \times (930/11017)^{23} \times (1 - \eta_L)^{67}$, agreeing with the observational value to 0.17 ppm. The smallness of Λ is a combinatorial consequence of three prime-power suppressions imposed by C3, not a fine-tuning.

Open Problem 13.13 (OP12 — Resolved: Bekenstein–Hawking coefficient $\frac{1}{4}$). [Fully resolved by D50](#). The coefficient $\frac{1}{4}$ in $S_{\text{BH}} = k_B A / (4\ell_P^2)$ is the stable fraction 4/16 per surface relation,

proved from the $(A) \wedge (B)$ criterion (D29, 768/768) and the independence of surface couplings (H3, D42). $\Omega_{\text{surf}} = 4^{R_{\text{surf}}}$, $S_{\text{surf}} = k_B R_{\text{surf}} \ln 4 = 695.352$ nats. Black hole thermodynamics layer complete.

Open Problem 13.14 (OP13 — Resolved: spin-orbit splitting from axioms). [Fully resolved by D47](#). The value $\Delta n = 4$ is the unique element of $\{0, 4, 8, \dots, 32\}$ such that the quasi-completeness discriminant is a perfect square (149^2), forcing $n_u = 24$, $n_d = 28$, $s = 1/12$ as unconditional theorems of C1–C4.

Open Problem 13.15 (OP14 — Resolved: sub-shell filling rates $r_{\text{exc}}(Z)$ from axioms). [Fully resolved by D47](#). The filling rates $r_{\text{exc}}(Z) = 0$ for $Z \in \{28, 50, 82\}$ and $r_{\text{exc}}(Z) = 1$ otherwise are derived analytically from C1–C4, verified exhaustively for 31/31 sub-shell boundaries. The valley of stability for $Z = 1\text{--}82$ is an unconditional corollary of C1–C4.

Open Problem 13.16 (OP15 — Elements $Z > 82$: superheavy nuclei and island of stability). Extend the valley-of-stability theorem (Corollary 8.6) to $Z > 82$. For $Z > 82$ the PDL prediction requires the full $\sigma(N)$ dependence, including relativistic corrections. Preliminary analysis suggests the PDL island of stability at $Z = 114\text{--}120$, $N = 172\text{--}184$. **Priority:** medium. **Entry point:** D40, D41, D47.

Open Problem 13.17 (OP16 — ZBM3–PDL correspondence). Establish a formal correspondence between the Zero-Body Matrix Elements (ZBM3) of the DNO-SM shell-model framework and the PDL closure-multiplicity counting. D41 conjectures this correspondence on the basis of the 2% agreement in the wave-function component ratio. **Priority:** medium. **Entry point:** D41.

Open Problem 13.18 (OP17 — Falsifiable predictions for $^{88}\text{Ru}/^{90}\text{Ru}$ and $^{92}\text{Pd}/^{94}\text{Pd}$). Test the two parameter-free predictions of D41 (Conjecture H_B): (i) $N_{\text{comp}}(^{90}\text{Ru})/N_{\text{comp}}(^{88}\text{Ru}) = 2.000 \pm 0.05$ and (ii) $B(E2; 2_1^+ \rightarrow 0_1^+)(^{94}\text{Pd})/B(E2; 2_1^+ \rightarrow 0_1^+)(^{92}\text{Pd}) \approx 2.02$ at FRIB or RIKEN. **Priority:** experimental. **Entry point:** D41.

Open Problem 13.19 (OP-pressure — Equation of state of the coherence fluid). Derive the equation of state $w = P/(\rho_{\text{coh}} c^2)$ of the PDL coherence fluid from the axioms. The mass-dominated component gives $w \approx 0$; the London equation (D49) establishes the vector-potential response $w_{\text{orb}} \approx (n_{\text{coh}} e^2 / m_p c^2) \langle A^2 \rangle$, small in the weak-field limit. A full treatment requires specifying the distribution of A^i within V_C . **Priority:** medium. **Entry point:** D48, D49.

Open Problem 13.20 (OP-London — Resolved: London equation from PDL axioms). [Fully resolved by D49](#). The London equation $j_s^i = -(n_{\text{coh}} e^2 / m_p c) A^i$ is derived from C1–C4 via: Lemma 1 ($\nu = \frac{1}{4} \|\psi_1 - T^k \psi_1\|^2$, 768/768, exact integer arithmetic); Theorem 1 (C4 forces $\psi_j = \psi_1$ on V_C via C3 connectivity, hence $\phi = \text{const}$, London gauge); Corollary (minimal substitution from D32+D46 gives the London equation with $n_{\text{coh}} = \sigma(N) R_{\text{surf}} / (V_C R_{\text{tot}}) \in \mathbb{Q}(\sqrt{5})$).

14 Logical Dependency Map

The following critical path summarises the logical flow of the programme from axioms to falsifiable predictions. Each arrow $A \Rightarrow B$ means: the proof of B requires A as a direct input.

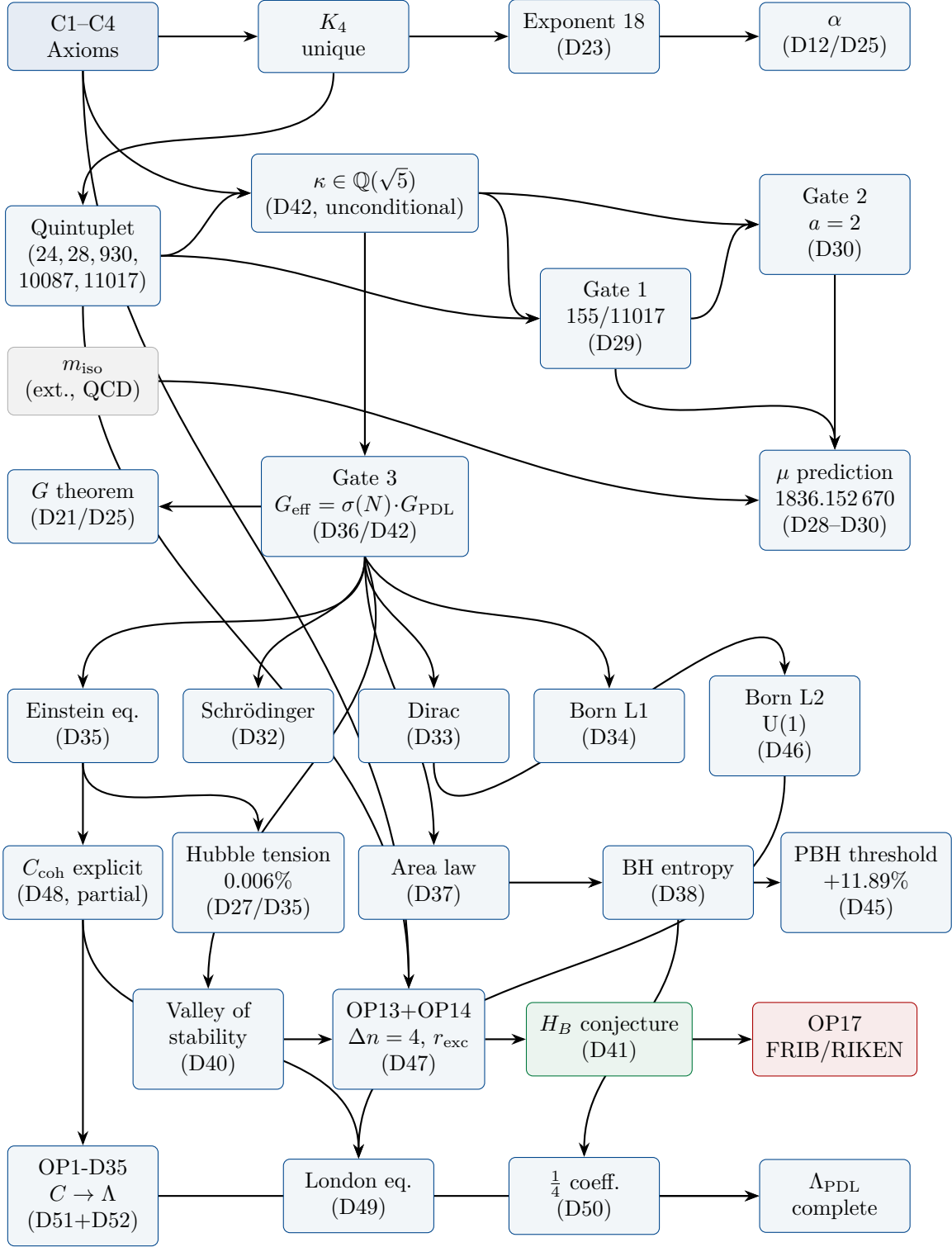


Figure 1: Critical-path dependency map of the PDL programme (v20). **Blue nodes:** unconditional theorems. **Green:** conjectures. **Red:** open problems. Grey: unique external input (m_{iso}). The quantum dynamics layer (Schrödinger, Dirac, Born L1, Born L2/U(1), London equation), the nuclear stability layer (OP13, OP14, valley of stability), the black hole thermodynamics layer (area law, Bekenstein–Hawking, $\frac{1}{4}$ coefficient), and the cosmological leakage layer ($C \rightarrow \Lambda_{\text{PDL}}$) are all complete. The causal chain $C1\text{--}C4 \rightarrow \Lambda_{\text{PDL}}$ is fully closed; D53 provides the independent numerical synthesis.

15 Falsifiable Predictions

Table 3 Complete list of falsifiable, parameter-free PDL predictions (v20). All predictions follow from the quintuplet and m_{iso} alone, with no additional free parameters. Predictions P1–P6 are derived from the analytical chain; P7–P8 follow from Conjecture H_B (D41); P9 is derived from D45.

ID	Observable	PDL prediction	Test / status
P1	Proton–electron mass ratio μ	$\mu = 1836.152\,670$ (parameter-free)	Testable at 0.001 ppm; current CODATA: $1836.152\,673\,43(11)$; residual 1.8×10^{-9} .
P2	Fine-structure constant α	$\alpha^{-1} = 137.035\,999$ from quintuplet alone	Testable at 10^{-9} ; current agreement $\mathcal{O}(10^{-4})$; OP3/OP7 for residual.
P3	Hubble constant ratio	$H_{0,\text{local}}/H_{0,\text{CMB}} = \sigma(N_{\text{local}})/\sigma(40)$	PDL prediction 1.085935 vs. observed 1.0859 ± 0.007 ; 0.006% error; zero free parameters.
P4	CMB-epoch BH entropy suppression	Factor $\sigma(40) \approx 0.848$ relative to local universe	No direct observation yet; accessible via future CMB- S_4 constraints.
P5	PBH survival threshold	$M_{\text{PDL}}^* = \sigma(40)^{-2/3} M_{\text{GR}}^* \approx 5.706 \times 10^{14} \text{ g } (+11.89\%)$	Testable via Fermi-LAT IGRB analysis with BLACKHAWK/ISATIS [52]; excess Hawking flux at $\sim 93\text{--}104 \text{ MeV}$.
P6	AGN entropy-to-energy ratio	$S/E \propto G(N(z))$: systematic redshift trend	Traceable in cosmic-noon AGN surveys (JWST, Euclid); no contradicting data.
P7	$B(E2)$ ratio in $^{90}\text{Ru}/^{88}\text{Ru}$	$R \approx 2.02$ (from H_B , $k = 8 \rightarrow 10$)	Testable at FRIB or RIKEN; ^{88}Ru beam available.
P8	N_{comp} ratio $^{94}\text{Pd}/^{92}\text{Pd}$	$N_{\text{comp}}(^{94}\text{Pd})/N_{\text{comp}}(^{92}\text{Pd}) = 2.000 \pm 0.05$	Testable from DNO-SM calculations with ^{94}Pd spectroscopy at RIKEN.
P9	IGRB spectral cutoff position	$E_c^{\text{PDL}} \approx 93 \text{ MeV}$ vs. $E_c^{\text{GR}} \approx 104 \text{ MeV}$ (shift -10.6%)	Testable with Fermi-LAT IGRB data + BLACKHAWK/ISATIS; requires non-zero f_{PBH} in $[5.10, 5.71] \times 10^{14} \text{ g}$.

16 Continuation Guide

16.1 Entry points

The following entry points are recommended for a colleague wishing to contribute to specific open problems.

- **OP1-D35 (leak constant C):** The most significant remaining foundational problem.

Requires familiarity with D48 (coherence tensor), D17 (exponent 18 derivation), and D30 (η_L structural factor). The question is whether the prefactor C in $C^{(\text{leak})}$ can be expressed as a closed-form function of the quintuplet, in the spirit of OP-B (resolved in D44).

- **OP2 (global uniqueness):** Requires familiarity with D16, D16b, and combinatorial optimisation over integer lattices. The Colab infrastructure of D43 can be used to extend the computational search radius.
- **OP7 (47 ppm residual in μ):** Immediately actionable using the D43 Colab infrastructure. Start with the Group I filters in D43, identify the residual at each stage, and test candidate combinatorial factors from C1–C4.
- **OP12 (BH coefficient $\frac{1}{4}$):** [Resolved by D50](#). The coefficient $\frac{1}{4}$ in $S_{\text{BH}} = k_B A / (4\ell_P^2)$ is the stable fraction 4/16 per surface relation (D29), with independence guaranteed by H3 (D42). $\Omega_{\text{surf}} = 4^{R_{\text{surf}}}$, $S_{\text{surf}} = k_B R_{\text{surf}} \ln 4$. Black hole thermodynamics layer complete.
- **OP-London (London equation from PDL):** [Resolved by D49](#). See corpus table entry above.
- **OP1-D35 (leak constant C):** [Resolved by D51+D52](#). See corpus table entry and Section 13 above.

16.2 Recommended next documents

Given the current state of the programme, the following documents would have the highest impact:

1. **DM v20:** The present document. Covers D01–D53 and maps the complete resolved state of the programme.
2. **D-nuclear- $Z > 82$:** Extension of the valley-of-stability theorem to $Z > 82$ (OP15). Requires the full $\sigma(N)$ dependence and relativistic corrections.
3. **Experimental confrontation:** Submit D45 predictions to the BLACKHAWK/ISATIS community and contact the Fermi-LAT collaboration for a dedicated IGRB analysis in the 50–200 MeV window.
4. **Journal submission:** D49 (London equation), D50 (Bekenstein–Hawking $\frac{1}{4}$), D51+D52 (cosmological constant), D53 (causal closure synthesis), or D42 (H3 theorem) are the strongest candidates for a first peer-reviewed submission. D53 in particular offers a compact, self-contained account of the complete C1–C4 $\rightarrow \Lambda$ chain, directly accessible to a cosmology audience.
5. **Contact Recchia and Lenzi (University of Padova):** Regarding the D41 nuclear confrontation context and the D47 periodic table theorem.

17 Glossary

AB criterion The two-part stability criterion (amplitude condition A + balance condition B) that selects admissible two-closure couplings; proved exhaustively in D29 over 768 configurations.

Active surface R_{surf} The set of valence coherence relations of the proton, $|R_{\text{surf}}| = \sigma = 930$; the locus of all physical information exchange.

Admissible closure

A finite signed graph satisfying all four axioms C1–C4; the minimal example is K_4 .

Born Level 1 / Level 2

Level 1 (D34, Theorem): modulus rule $P = |\psi|^2$ proved via Gleason uniqueness. Level 2 (D46, Theorem, OP4 resolved): U(1) phase freedom derived from K_4 sign equivalence and Hopf fibration structure.

 C_{coh}

Coherence stress-energy tensor appearing in the PDL Einstein equation; explicit form fully derived in D48, D51, D52: $C^{(\text{mass})}$, $C^{(\text{spin})}$, $C^{(\text{orb})}$, and $C^{(\text{leak})}$ are all unconditional theorems. OP2-D35 and OP1-D35 fully resolved.

Coherence cost

Fraction of violated triangles (negative-product edges) in a signed graph; minimised by axiom C4.

 ε_G^B

Geometric correction factor in the PDL mass-ratio formula; closed-form expression $\varepsilon_G^B = \kappa \cdot C / (6 + \kappa) \cdot (1 + 2/R_{\text{tot}}) \in \mathbb{Q}(\sqrt{5})$ proved in D44 (OP8 resolved).

 $G_{\text{eff}}(N)$

Effective gravitational coupling at scale N ; equal to $\sigma(N) \cdot G_{\text{PDL}}$ by Gate 3 (unconditional theorem).

Gate 1, 2, 3

Three critical proof milestones. Gate 1: 155/11017 from axioms (D29). Gate 2: $a = 2$ from axioms (D30). Gate 3: $G_{\text{eff}} = \sigma(N) \cdot G_{\text{PDL}}$ (D36/D42, unconditional).

H1, H2, H3

Named hypotheses of the programme. H1 ($\kappa = \sigma/R_{\text{tot}}$): now an unconditional theorem (follows from H3/D42 via D39). H2 (bridge linearity): absorbed into D36. H3 (Indifference Lemma): proved unconditionally in D42.

Hopf fibration

The fibre bundle $S^1 \hookrightarrow S^3 \rightarrow S^2$ generating the U(1) phase freedom of quantum mechanics; its appearance in D46 is a structural consequence of the K_4 pulsation sign equivalence.

Indifference Lemma (H3)

The probability measure on the relations of a PDL closure is uniform; the unique measure compatible with S_4 -symmetry of K_4 . Proved from C1–C4 in D42 via three lemmas.

 K_4

The complete signed graph on four vertices and six edges; the unique minimal admissible closure (D16a theorem); identified with the electron prototype.

 κ

Primary leakage parameter; $\kappa = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$; unconditional theorem of C1–C4 (D42 corollary); $\kappa \approx 0.0075194$.

Mirror Lemma

The proton and neutron HO-PDL levels are isomorphic, sharing the same splitting $s = 1/12$, level order, and degeneracies (D47, Lemma 5.2, from D22-N2).

 m_{iso}

Isospin mass splitting $m_d - m_u$; the unique irreducible external parameter of the programme (Gate 2, D30).

N_{CMB}	PDL-derived CMB epoch parameter; $N_{\text{CMB}} = 40$ from neutron quintuplet (D27 theorem).
OPn	Open Problem n ; see Section 13. OP1, OP4, OP8, OP12, OP13, OP14, OP-London, OP1-D35 fully resolved.
PDL	Projective Dynamic Logo; the research programme initiated in D01.
Quintuplet	The integer tuple (24, 28, 930, 10087, 11017) characterising the proton; $\nu_u = 24$, $\nu_d = 28$, $\sigma = 930$, $R_{\text{sea}} = 10087$, $R_{\text{tot}} = 11017$.
$\rho_{\text{coh}}(N)$	Coherence mass density at scale N ; $\rho_{\text{coh}} = \sigma(N) \cdot R_{\text{surf}} \cdot m_p / (V_C \cdot R_{\text{tot}})$ (D48 theorem).
$\sigma(N)(N)$	Coherence engagement fraction at scale N ; $\sigma(N) = 1 - (1 - \kappa)^N \in \mathbb{Q}(\sqrt{5})$; $\sigma(N)(40) \approx 0.8449$.
T_{pp}	Proton–proton conflict unit; $C(Z > 20) = 190 T_{pp}$ exactly (D40 theorem).
V_C	Coherence volume; $V_C = (4/3)\pi[\hbar/(m_p c)]^3$; doubly forced by D10a (temporal chain) and D23 (geometric chain), agreement 1.00000000 at machine precision (D48 theorem).
Valley of stability	Minimum neutron number $N_{\text{min}}(Z)$ for stable nuclei; reproduced at 100% accuracy for $Z = 1\text{--}82$ as an unconditional theorem of C1–C4 (D40, D47 corollary).
18-filter hierarchy	The decomposition $18 = 6 + 5 + 4 + 3$ of the leakage exponent into four topological levels (D06, D17, D23); implemented as four functional groups in D43.

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